

From Static Retrieval to Active Cognition: A Review of Learnable Memory Architectures for Long-Horizon LLM Agents

The deployment of Large Language Model (LLM) agents in long-horizon tasks is fundamentally constrained by finite context windows, creating a "memory bottleneck" that leads to context explosion and semantic drift [16]. This review synthesizes the paradigm shift from passive, static Retrieval-Augmented Generation (RAG) toward active, learnable memory management strategies. We analyze how traditional approaches fail to resolve contradictions or maintain causal consistency, often resulting in the "lost in the middle" phenomenon and faulty memory consolidation [75]. In response, emerging architectures treat memory operations as integral components of the agent's policy, utilizing Reinforcement Learning (RL) to optimize what, when, and how information is stored or discarded [10]. Key trends include hierarchical structures separating episodic and semantic layers, indexed experience memory for lossless evidence preservation, and graph-based representations for dependency-structured reasoning [7; 18]. Despite advances in biologically inspired forgetting and active context curation, significant challenges remain regarding credit assignment in multi-session settings and the reliability of continuous abstraction. We conclude that achieving robust, generalizable agency requires moving beyond simple retrieval to dynamic, self-correcting memory systems that balance fidelity, efficiency, and causal coherence.

Introduction: The Memory Bottleneck and the Shift to Active Management

The deployment of Large Language Model (LLM) agents in long-horizon tasks is fundamentally constrained by finite context windows, creating a critical "memory bottleneck" that leads to context explosion, semantic drift, and the loss of vital evidence required for multi-step reasoning. While modern architectures support increasingly large contexts, relying solely on full-context prompting or static Retrieval-Augmented Generation (RAG) has proven inefficient due to attention dilution, the "lost in the middle" phenomenon, and prohibitive computational costs [16]. The core limitation lies in the stateless nature of standard LLMs, which struggle to maintain behavioral consistency or track knowledge evolution over extended interactions without explicit, managed memory structures [276]. Consequently, the field is witnessing a paradigm shift from passive, heuristic-driven memory systems toward active, learnable memory management strategies that treat memory operations as integral components of the agent's policy.

Traditional approaches often treat long-term memory (LTM) and short-term memory (STM) as separate components governed by rigid rules or auxiliary controllers, limiting adaptability and preventing end-to-end optimization [10]. Static RAG pipelines, while popular, function essentially as stateless lookup tables lacking lifecycle control, versioning, or the ability to resolve contradictions, rendering them inadequate for dynamic, multi-turn workflows [109]. This structural gap necessitates a transition to "Continuum Memory Architectures" that maintain internal state through persistent storage, selective retention, and consolidation into higher-order abstractions, rather than treating memory as a transient retrieval event [14]. The emergence of frameworks like MemOS illustrates this systems-level perspective, proposing a unified operating system that manages plaintext, activation-based, and parameter-level memories as schedulable resources, thereby enabling controllability, plasticity, and evolvability [109].

A dominant trend in addressing these limitations is the integration of Reinforcement Learning (RL) to train agents in active memory curation. Instead of relying on pre-defined instructions for memory updates, recent frameworks empower agents to autonomously decide what, when, and how to store, retrieve, update, or discard information. For instance, Agentic Memory (AgeMem) unifies LTM and STM management directly into the agent's policy by exposing memory operations as tool-based actions, optimized through a progressive RL strategy [10]. Similarly, Memory-R1 employs specialized agents fine-tuned with outcome-driven RL to learn structured operations such as ADD, UPDATE, and DELETE, demonstrating that adaptive memory management can be achieved with minimal supervision [149]. This move toward learned policies extends to context compression, where frameworks like MemAct treat working memory management as learnable policy actions, utilizing in-place editing operations to jointly optimize information retention and task performance [13].

The inefficiency of static retrieval has also spurred innovations in indexed and hierarchical memory structures that preserve evidence fidelity while managing context length. Methods like Memex introduce an indexed experience memory mechanism that compresses the working context into concise summaries while archiving full-fidelity interactions externally, allowing agents to dereference specific indices only when necessary [7]. This approach contrasts with lossy summarization techniques by ensuring that critical evidence is not discarded but rather offloaded to a structured store. Hierarchical approaches further refine this by organizing memory into layers of abstraction; HIAGENT leverages subgoals as memory chunks to manage working memory hierarchically, while H-MEM utilizes index-based routing for efficient layer-by-layer retrieval [1; 110]. These architectures address the "context rot" observed in flat retrieval systems by maintaining logical dependencies and causal structures essential for complex reasoning [18].

Furthermore, the distinction between procedural and episodic memory is gaining traction as a means to enhance agent adaptability. While episodic memory captures specific interaction traces, procedural memory distills these trajectories into reusable scripts and instructions, allowing agents to generalize across tasks [242]. Frameworks like REMEMBERER and Mem-? push this further by modeling memory as a generative policy or a semi-parametric system, where the agent learns to synthesize guidance on demand rather than retrieving static entries, thus avoiding the misalignment often found in similarity-based retrieval [276; 162]. The consensus across these diverse approaches is clear: effective long-horizon agency requires a departure from static, external memory banks toward dynamic, internalized, and learnable memory mechanisms that actively curate the agent's cognitive state [117]. As benchmarks like AgentLongBench and EvoMemBench reveal, current systems still struggle with dynamic information synthesis,

underscoring the necessity for these active management strategies to bridge the gap between static retrieval and genuine lifelong learning [76; 204].

The Limitations of Static Context and Passive Retrieval

Traditional approaches to equipping Large Language Model (LLM) agents with long-term memory have predominantly relied on two paradigms: full-history concatenation within expanding context windows or standard Retrieval-Augmented Generation (RAG) based on semantic similarity. While these methods offer a straightforward implementation path, they exhibit profound brittleness when confronted with the dynamic, evolving nature of real-world interactions. The fundamental flaw in static context management is the assumption that all historical tokens possess uniform utility or that semantic proximity equates to causal relevance. As interaction horizons extend, naive concatenation leads to "context rot," where early errors, outdated facts, and irrelevant conversational filler obscure critical information, causing performance degradation known as the "Lost in the Middle" phenomenon [11; 44]. This accumulation of low-entropy noise not only inflates computational costs but actively degrades the model's reasoning capabilities by diluting the signal-to-noise ratio within the attention mechanism [63; 278].

The limitations of passive retrieval are even more pronounced when dealing with conflicting or updated information. Standard RAG systems typically treat memory as a stateless lookup table, retrieving chunks based on vector similarity without regard for temporal validity or logical consistency. This approach fails to distinguish between a user's past preference and their current, revised intent, leading to the retrieval of stale or contradictory data that misguides the agent's response [190; 40]. Research indicates that semantically similar memories can be causally irrelevant or even harmful if they represent superseded states of the world. For instance, in domains requiring precise state tracking, such as scientific research or software engineering, retrieving a parameter setting from an earlier experimental phase that has since been invalidated can lead to catastrophic reasoning errors [11; 22]. The inability of passive systems to perform causal interventions or assess the "usefulness" of a memory beyond its semantic overlap with a query results in a fragile architecture that struggles to maintain coherence over long horizons [40].

Furthermore, the reliance on static retrieval mechanisms often fragments the narrative and causal flow of interactions. By breaking dialogue streams into isolated utterances or fixed-size chunks for embedding, traditional systems irreversibly damage the structural dependencies required for multi-hop reasoning [27; 31]. This fragmentation prevents agents from reconstructing the execution state necessary to understand how one decision influenced subsequent outcomes, a capability essential for complex, interdependent tasks. Consequently, agents may retrieve factually correct snippets that are logically disconnected from the current reasoning path, leading to hallucinations or incoherent responses [183; 294]. The problem is exacerbated by the fact that current consolidation techniques, which attempt to compress history into summaries, often introduce their own forms of distortion. Studies have shown that iterative summarization can lead to "faulty memory," where useful details are dropped, spurious rules are introduced, and the consolidated abstraction drifts away from the underlying task structure, sometimes performing worse than having no memory at all [75; 70].

The inadequacy of these static and passive approaches is further highlighted by their failure to manage interference in multi-target scenarios. When information is repeatedly updated, as is common in long-running agent deployments, existing benchmarks and systems struggle to recall earlier facts that have been revised by subsequent context [44]. The retrieval process in standard RAG does not inherently possess a mechanism to resolve conflicts between retrieved documents, leaving the burden of contradiction resolution entirely to the LLM's context window, which is already saturated with noise. This creates a bottleneck where the agent cannot effectively differentiate between a persistent fact and a transient observation, leading to inconsistent behavior and a lack of trustworthiness in long-term deployments [82; Hierarchical Long-Term Semantic Memory for LinkedIn's Hiring Agent]. Ultimately, the shift from passive retrieval to active memory management is necessitated by the realization that memory is not merely a storage problem but a dynamic reasoning challenge that requires continuous evaluation of relevance, consistency, and causal utility [14; 109].

The Paradigm Shift to Active and Learnable Memory

The field of autonomous agent development is undergoing a fundamental transformation, moving away from static, database-like memory architectures toward dynamic frameworks where memory is treated as an executable component managed by learned policies. Historically, agents relied on passive retrieval mechanisms or simple sliding windows, approaches that often fail under the pressure of long-horizon tasks due to "context bloat" and the dilution of attention [177]. As interaction histories expand, computational costs explode and reasoning capabilities degrade, necessitating a decoupling of the working context size from the total interaction history. This shift enables agents to actively curate, compress, and revise their working context, adapting their memory strategies based on the evolving demands of the task rather than adhering to fixed heuristics [38].

A central tenet of this new paradigm is the transition from passive storage to active context management, where the agent itself decides what information to retain, summarize, or discard. Systems like Focus demonstrate that agents can autonomously self-regulate their context by consolidating key learnings into persistent knowledge blocks while pruning raw interaction history, achieving significant token reduction without sacrificing accuracy [177]. Similarly, AgentFold introduces a proactive "folding" operation inspired by human cognitive consolidation, allowing agents to sculpt their context dynamically at multiple scales, from granular condensation to deep abstraction of sub-tasks [58]. This approach contrasts sharply with traditional Retrieval-Augmented Generation (RAG) systems, which often treat memory as a stateless lookup table lacking temporal continuity or the ability to mutate stored information [14]. By treating context as a dynamic cognitive workspace, these active systems maintain high information density, ensuring that decision-relevant information is preserved within finite context budgets [38].

Crucially, the management of this memory is increasingly being framed as a learnable policy rather than a hard-coded rule set. Recent research posits that memory operations—such as extraction, consolidation, and pruning—should be reframed as learnable and evolvable skills. MemSkill, for instance, employs a controller that learns to select relevant memory skills and evolves the skill set itself through a closed-loop procedure that reviews hard cases [47]. This aligns with the broader concept of "Memory as Action," where context management is formulated as in-place editing operations optimized through end-to-end reinforcement learning [13]. Frameworks like AtomMem further deconstruct high-level memory processes into atomic CRUD (Create, Read, Update, Delete) operations, transforming the memory workflow into a dynamic decision-making problem that allows agents to discover structured, task-aligned management strategies [66]. Such learnable approaches address the rigidity of static, hand-designed operations that often fail to adapt to diverse interaction patterns [47].

Despite the promise of active management, significant challenges remain regarding the reliability of memory consolidation and the potential for information degradation. While many systems aim to distill trajectories into reusable abstractions, research indicates that continuous updates by LLMs can introduce faults, causing memory utility to degrade over time as spurious rules are introduced or applicability conditions are lost [75]. This finding suggests a critical tension between the need for compression and the risk of losing evidentiary fidelity. To mitigate this, some architectures propose separating raw episodic evidence from consolidated abstractions, treating raw episodes as first-class evidence while gating consolidation explicitly [75]. Others, like Memex, utilize an indexed experience memory that compresses the working context into summaries while archiving full-fidelity artifacts externally, allowing for precise dereferencing when specific evidence is required [7]. This hybrid approach attempts to balance the efficiency of compression with the robustness of retaining original data, a strategy also seen in systems that employ hierarchical memory orchestration to surface historical knowledge precisely when needed [160].

Furthermore, the integration of biological principles into memory control is gaining traction as a method to handle the complexities of forgetting and retention. Frameworks such as OBLIVION and FadeMem incorporate decay-driven activation and biologically inspired forgetting mechanisms, respectively, to manage memory accessibility and prevent interference [48; 291]. These systems recognize that selective forgetting is as critical as retention for maintaining stable multi-turn behavior and preventing unbounded context growth [61]. The emergence of benchmarks like EvoMemBench and LongMemEval-V2 underscores the community's focus on evaluating these dynamic capabilities, moving beyond static recall to assess how agents internalize environment-specific experience and evolve over time [204; 43]. Ultimately, the shift toward active and learnable memory represents a move from viewing memory as a passive

repository to recognizing it as a core, adaptive module that governs the agent's ability to reason, plan, and execute complex tasks over extended horizons [16].

Hierarchical and Multi-Layered Memory Architectures

To address the critical bottleneck of context window saturation in long-horizon tasks, contemporary research has increasingly converged on decomposing agent memory into distinct, hierarchical layers that mimic human cognitive structures. This paradigm shift moves away from monolithic context accumulation toward architectures that separate fast-changing episodic evidence from slow-varying semantic knowledge, thereby balancing immediate reasoning needs with long-term retention. A foundational approach in this domain is the Dual-Process Memory Architecture, which decouples a constant-sized episodic buffer for immediate interactions from a growing neocortical memory for consolidated knowledge. This separation allows agents to maintain high accuracy over thousands of messages by preventing context overflow while ensuring that domain-specific contradictions and parameter evolutions are resolved through principled consolidation rather than uniform aggregation [11]. Similarly, the HINDSIGHT framework organizes memory into four logical networks distinguishing world facts, agent experiences, entity summaries, and evolving beliefs, treating memory as a structured substrate for reasoning rather than a passive retrieval layer [54].

The implementation of these hierarchies often draws inspiration from operating system designs and cognitive theories of engram formation. For instance, EverMemOS implements a self-organizing lifecycle where episodic traces are converted into atomic "MemCells," which are subsequently consolidated into thematic "MemScenes" to distill stable semantic structures [42]. This multi-stage process ensures that retrieval is guided by high-level semantic scenes, allowing the agent to reconstruct necessary context without loading the entire history. Complementing this, HiMem constructs a hierarchical structure linking concrete Episode Memory with abstract Note Memory, utilizing a Topic-Aware Event-Surprise segmentation strategy to bridge interaction events with stable knowledge [33]. These architectures collectively argue that separating temporal dynamics from semantic stability is essential for scalability, as evidenced by the failure of full-context models at the "Cognitive Event Horizon" where noise obscures critical facts [11].

A significant divergence in the literature exists regarding the mechanism of consolidation and the role of the LLM in managing these layers. Some systems, such as MemGPT, employ explicit LLM-driven memory management where the model must learn to invoke paging functions, effectively requiring meta-cognitive reasoning about its own memory constraints [11]. In contrast, other approaches advocate for implicit, automatic consolidation where background processes handle memory updates independently of the primary inference path to ensure robustness [11]. Furthermore, the timing of consolidation remains a point of contention; while many systems consolidate after every interaction, RecMem proposes a recurrence-based strategy where consolidation is triggered only when sustained semantic similarity is observed, significantly reducing token consumption [45]. This debate highlights a trade-off between the flexibility of agent-controlled paging and the stability of automated, asynchronous consolidation.

Beyond the episodic-semantic dichotomy, recent works introduce working memory as a distinct, managed layer to handle subgoals and immediate task states. HIAGENT leverages subgoals as memory chunks to hierarchically manage working memory, allowing agents to replace previous subgoals with summarized observations and retain only action-observation pairs relevant to the current objective [1]. This approach mirrors human problem-solving strategies by focusing attention on current sub-tasks while archiving completed trajectories. Similarly, CogMem incorporates a Focus of Attention mechanism that dynamically reconstructs concise, task-relevant context at each turn, sitting atop layers for direct access and long-term memory to mitigate reasoning drift [116]. The integration of these layers is further refined in frameworks like TiMem, which utilizes a Temporal Memory Tree to organize conversations from raw observations to abstracted persona representations, treating temporal continuity as a first-class organizing principle [36].

Despite the consensus on the utility of hierarchy, challenges remain regarding the fidelity of information during the abstraction process. Critics note that aggressive consolidation can lead to "faulty memory" where useful details are dropped or spurious rules introduced, causing performance to degrade below baseline levels over time [75]. To counteract this, architectures like Memex introduce indexed experience memory, which maintains a compact working context of summaries while preserving full-fidelity underlying interactions in an external archive, allowing for explicit dereferencing when granular evidence is required [7]. This hybrid approach attempts to resolve the tension between the efficiency of abstraction and the necessity of precise evidence retention. Additionally, systems like MEMORA and SYNAPSE explore harmonic representations and spreading activation mechanisms to balance specificity with

abstraction, ensuring that fine-grained details remain accessible through associative links even as higher-level summaries are formed [77][67].

The evolution of these architectures also reflects a move toward dynamic, adaptive forgetting and gating mechanisms. CraniMem and FadeMem incorporate biologically inspired forgetting processes, using gated mechanisms and adaptive decay rates to prune low-utility items and prevent interference from distractor content [137][291]. Such mechanisms are crucial for maintaining a bounded state within an infinite horizon, ensuring that memory growth does not become unmanageable. Furthermore, the integration of reinforcement learning, as seen in Mem-? and Memory-R1, enables agents to learn optimal policies for when to store, update, or discard information, moving beyond static heuristics toward end-to-end optimized memory management [89][149]. Collectively, these diverse approaches underscore that effective long-horizon reasoning requires not just larger context windows, but sophisticated, multi-layered architectures that actively govern the flow, consolidation, and retrieval of information across varying temporal scales.

Working Memory Optimization via Subgoals and Summarization

Effective management of working memory in Large Language Model (LLM) agents is critical for sustaining coherent reasoning over long-horizon tasks, where the accumulation of raw action-observation pairs often leads to context saturation and performance degradation. A primary strategy to mitigate this issue involves the compression of historical data into concise subgoals or summaries that retain only task-relevant information. The HIAGENT framework exemplifies this approach by leveraging subgoals as hierarchical memory chunks, prompting agents to proactively replace outdated details with summarized observations relevant to the current subgoal. This method has been shown to double success rates and significantly reduce the number of steps required in complex tasks compared to naive history concatenation [1]. Similarly, the MemoBrain framework operates as an executive co-pilot that actively manages the working context by pruning invalid steps and folding completed sub-trajectories, thereby preserving a compact, high-salience reasoning backbone under fixed context budgets [34]. These systems shift the paradigm from passive context accumulation to explicit cognitive control, ensuring that logical continuity is maintained without overwhelming the model's attention mechanisms.

The implementation of subgoal-based compression often relies on hierarchical structures that mirror human problem-solving strategies. For instance, MAGE (Memory as Agent-Guided Exploration) organizes interactions within a hierarchical state tree, where completed subgoals are compressed while maintaining the integrity of the active root-to-current path. This design effectively isolates flawed segments from the active trajectory, bounding context growth while preserving state integrity [31]. In a similar vein, ReAcTree decomposes complex goals into manageable subgoals within a dynamically constructed agent tree, where each node handles specific sub-tasks and shares environment-specific observations through working memory. This hierarchical decomposition allows agents to avoid the entanglement of all past decisions in a monolithic trajectory, leading to substantial improvements in goal success rates [218]. Furthermore, the concept of "context folding" introduced by AgentFold treats the context as a dynamic workspace where agents learn to execute granular condensations or deep consolidations of multi-step sub-tasks. This proactive management enables agents to maintain highly focused contexts even after hundreds of interaction turns, matching the performance of significantly larger models [58].

While subgoal abstraction is powerful, it introduces the risk of information loss if critical details are discarded during compression. To address this, several frameworks propose hybrid or indexed approaches that balance abstraction with fidelity. Memex introduces an indexed experience memory mechanism that compresses the working context into structured summaries while archiving full-fidelity interactions in an external database under stable indices. This allows agents to dereference specific indices to recover exact past evidence when needed for a current subgoal, effectively compressing context without discarding evidence [7]. Complementing this, the M2 framework utilizes a dual-tier memory mechanism that synergizes dynamic trajectory summarization for internal memory with insight retrieval from an external bank, optimizing both context efficiency and decision-making robustness without requiring extensive retraining [50]. Additionally, SAM (State-Adaptive Memory) consolidates interactions into compact cues while preserving raw trajectory pages for intent-driven recall, allowing agents to reconstruct temporally distant information based on their evolving state [20].

However, the process of consolidation and summarization is not without its pitfalls. Research indicates that continuously updating consolidated memories can lead to faulty abstractions where useful details are dropped or spurious rules are introduced, sometimes causing performance to degrade below baseline levels [75]. This highlights the necessity of gating consolidation mechanisms and treating raw episodes as first-class evidence rather than immediately abstracting them. Consequently, frameworks like HyMem and CogMem adopt multi-granular or layered architectures. HyMem employs a dual-granular storage scheme with a lightweight module for summary-level context and a deep module activated only for complex queries, dynamically scheduling retrieval based on cognitive economy [189]. CogMem further refines this by incorporating a Focus of Attention mechanism that dynamically reconstructs concise, task-relevant context at each turn, mitigating reasoning failures and controlling context growth [116].

The efficacy of these optimization strategies is further validated by benchmarks designed to test long-term memory utilization. Evaluations on benchmarks such as LongMemEval-V2 and MEMORYARENA reveal that agents capable of distilling experiences into memory and proactively using them to guide future actions significantly outperform those relying on simple retrieval or full-context methods [43] [12]. Moreover, end-to-end reinforcement learning frameworks

like MemAct and AtomMem treat memory management as learnable policy actions, enabling agents to autonomously decide when to compress, delete, or update information, thereby adapting their strategies to specific task complexities [13] [66]. These advancements collectively demonstrate that effective working memory optimization requires a sophisticated interplay between hierarchical subgoal formulation, selective summarization, and the preservation of retrievable raw evidence to ensure robust performance in long-horizon agentic tasks.

Separation of Episodic and Semantic Layers

The architectural decoupling of episodic and semantic memory layers represents a fundamental shift in designing long-horizon agents, moving away from monolithic context windows toward structured systems that distinguish between raw interaction traces and consolidated knowledge. This separation is critical for maintaining stability in domains requiring precise fact retention and the management of evolving hypotheses, such as scientific research. The Dual-Process Memory Architecture exemplifies this approach by maintaining a fixed-size episodic buffer for recent context while incrementally consolidating long-term knowledge into a growing semantic store. This design ensures that agents can handle contradictory updates—such as shifting significance thresholds or pivoting biological hypotheses—without corrupting the core knowledge base, a capability that full-context models lack due to context overflow and cognitive degradation [11]. By isolating immediate episodic needs from long-term consolidated knowledge, these systems prevent the fragmentation of causal chains and support the emergence of stable user profiles that are robust to transient fluctuations.

The necessity of this separation is underscored by the fragility of continuous consolidation processes when episodic and semantic roles are collapsed into a single rewrite loop. Research indicates that useful memories often become faulty when continuously updated by large language models, as the consolidation step itself can introduce spurious rules or drop critical details, leading to performance degradation below no-memory baselines [75]. Consequently, robust architectures treat raw episodes as first-class evidence, gating consolidation explicitly rather than triggering it after every interaction. This aligns with findings that episodic-only controls, which retain raw trajectories without abstraction, often remain competitive with or superior to forced-consolidation counterparts in tasks requiring high fidelity [75]. Systems like EverMemOS implement this distinction through an engram-inspired lifecycle, where episodic trace formation captures atomic facts and time-bounded foresight before semantic consolidation organizes these traces into thematic structures, thereby resolving conflicts and distilling stable semantic structures [42].

Hierarchical frameworks further refine this separation by introducing intermediate layers that manage the flow of information between raw experience and abstract knowledge. Hierarchical Memory Orchestration (HMO) organizes interaction history into a three-tiered directory, coupling recent and pivotal memories with an evolving user profile to ensure reasoning remains aligned with individual behavioral traits while relegating less relevant information to global archives [160]. Similarly, HiMem constructs cognitively consistent episode memory via topic-aware segmentation and builds note memory for stable knowledge, linking these types semantically to bridge concrete events and abstract concepts [33]. This multi-layered approach addresses the "stability-plasticity dilemma," allowing agents to integrate transient interactions while preserving macroscopic user traits, a challenge that flat retrieval paradigms struggle to resolve due to their inability to manage cross-scale integration [53].

The separation of layers also facilitates more efficient context management and reduces the computational overhead associated with processing unbounded histories. MemForest reformulates agent memory as a write-efficient temporal data management problem, utilizing a hierarchical temporal index to organize memory as time-ordered trees rather than flat summaries, which allows for localized updates and preserves temporally evolving states without full-state rewrites [284]. In contrast, systems that fail to separate these layers often suffer from "context bloat," where irrelevant past errors distract the agent and increase latency. Focus, an agent-centric architecture, addresses this by autonomously deciding when to consolidate key learnings into a persistent knowledge block and actively pruning raw interaction history, demonstrating that capable models can self-regulate their context when given appropriate tools [177]. Furthermore, the distinction between episodic and semantic layers enables specialized retrieval strategies; for instance, dual-trace encoding pairs stored facts with concrete scene traces to improve cross-session recall and temporal reasoning, leveraging the specificity of episodic details to enhance semantic retrieval accuracy [65].

Despite the clear advantages of layered architectures, challenges remain in defining the boundaries and transition mechanisms between episodic and semantic stores. Some approaches, such as MemGPT, treat memory management as an operating system function where the model explicitly pages data between main context and external storage, yet this requires the LLM to learn meta-cognitive reasoning about memory operations, introducing potential failure modes [290]. In contrast, implicit automatic consolidation mechanisms decouple memory management from the primary inference path, offering greater robustness against consolidation failures [11]. The debate continues regarding the optimal granularity of consolidation, with some systems advocating for recurrence-based triggers that only invoke

consolidation when sustained patterns are observed, thereby reducing token consumption while maintaining accuracy [45]. Ultimately, the distinct separation of episodic and semantic layers is not merely an organizational preference but a structural necessity for agents operating in dynamic, long-horizon environments where the integrity of historical evidence must be preserved alongside the evolution of abstract knowledge.

Graph-Based and Structured Representations

Moving beyond the limitations of flat text storage and monolithic vector databases, graph-based memory representations have emerged as a critical paradigm for modeling complex relational structures among entities. These structured approaches facilitate multi-hop reasoning and the resolution of conflicting information through explicit edge connections, offering a robust alternative to standard vector similarity search which often fails to capture temporal dynamics and causal dependencies. Systems such as Mem0g and MAGMA utilize directed labeled graphs or orthogonal graphs to store memories, allowing for a deeper understanding of entity relationships and more resilient retrieval paths [9; 185]. By decoupling memory representation from retrieval logic, these architectures enable agents to navigate complex memory landscapes more effectively, supporting the aggregation of information across disparate sessions and outperforming baselines in tasks requiring factual consistency.

The efficacy of graph-based systems lies in their ability to disentangle different dimensions of memory. MAGMA, for instance, represents each memory item across orthogonal semantic, temporal, causal, and entity graphs, formulating retrieval as a policy-guided traversal over these relational views [185]. This contrasts with approaches like A-Mem, which draws upon the Zettelkasten method to dynamically organize memories into interconnected knowledge networks through agent-driven indexing and linking [217]. While A-Mem focuses on the dynamic evolution of links based on semantic similarity, other systems like SYNAPSE model memory as a dynamic graph where relevance emerges from spreading activation rather than pre-computed links, integrating lateral inhibition and temporal decay to filter interference [67]. Similarly, HeLa-Mem employs Hebbian learning dynamics within a dual-level organization, distinguishing between an episodic memory graph that evolves through co-activation and a semantic store populated via distillation, mirroring biological cognitive processes [148].

Hierarchical structures further enhance the scalability and precision of graph-based memory. GAAMA constructs a concept-mediated hierarchical knowledge graph with distinct node types for episodes, facts, reflections, and concepts, utilizing Personalized PageRank alongside semantic search to improve retrieval accuracy [286]. This hierarchical approach is echoed in GAM, which decouples memory encoding from consolidation by isolating ongoing dialogue in an event progression graph before integrating it into a topic associative network, thereby minimizing interference from transient noise [41]. Furthermore, H-MEM introduces a hybrid structure combining a temporal and semantic tree with a knowledge graph, allowing short-term data to evolve into summarized long-term memory while capturing entity relationships [210]. Such multi-layered designs address the "Contextual Tunneling" problem by enabling agents to traverse abstract summaries and concrete details seamlessly.

Despite the advantages of structured representations, contradictions exist regarding the optimal balance between static graph construction and dynamic reconstruction. While many systems rely on pre-defined graph extraction pipelines, MRAgent argues for an active reconstruction mechanism where memory retrieval is dynamically adapted to the reasoning context, integrating LLM reasoning directly into memory access to prune retrieval paths iteratively [294]. This challenges the rigid "retrieve-then-reason" paradigm, suggesting that static graphs may not suffice for highly dynamic environments. Additionally, concerns regarding the reliability of continuous graph updates have been raised; research indicates that useful memories can become faulty when continuously updated by LLMs, as consolidation steps may introduce spurious rules or overwrite critical evidence [75]. Consequently, systems like AriadneMem employ conflict-aware coarsening and algorithmic bridge discovery to resolve state updates and reconnect disconnected evidence without expensive iterative planning [96].

The integration of graph structures also supports specialized reasoning capabilities such as causal inference and procedural learning. ActMem transforms unstructured dialogue into a structured causal and semantic graph, leveraging counterfactual reasoning to resolve conflicts between past states and current intentions [198]. In parallel, Mem^{ap} explores the distillation of agent trajectories into procedural memory, organizing step-by-step instructions and script-like abstractions within a evolving repository [242]. These developments highlight a shift from viewing memory merely as a storage backend to treating it as an active, structured substrate for reasoning. However, the complexity of maintaining these graphs introduces engineering challenges, prompting lightweight alternatives like ENGRAM, which organizes conversation into canonical memory types using simple set operations rather than complex graph traversals, challenging the trend toward architectural complexity [182]. Ultimately, the literature suggests that while graph-based

representations offer superior capabilities for multi-hop and temporal reasoning, their implementation must carefully balance structural richness with computational efficiency and the risks of consolidation errors.

Adaptive Retrieval Gating and Retention Mechanisms

Advanced architectures for long-horizon dialogue systems increasingly rely on adaptive gating mechanisms to dynamically regulate the flow of information between memory layers, addressing the critical challenges of semantic drift and unbounded context growth. These systems move beyond static retrieval by implementing retention regularization and importance-weighted strategies that determine what information is retained, forgotten, or promoted based on real-time relevance scores. A foundational approach involves decomposing dialogue history into distinct working, episodic, and semantic layers, where adaptive retrieval gating controls cross-session drift while maintaining computational efficiency. Empirical evaluations on benchmarks such as LOCOMO and LoCoMo demonstrate that such multi-layered frameworks can achieve significant improvements in success rates and multi-hop F1 scores while drastically reducing false memory rates and context usage [5]. This structured approach contrasts with uncontrolled accumulation, which often leads to temporal decay and performance degradation in extended conversational settings.

To further mitigate the risks of information overload and false memory propagation, researchers have developed adaptive budgeted forgetting frameworks that regulate memory through relevance-guided scoring. By integrating factors such as recency, frequency, and semantic alignment, these systems maintain stability under constrained context budgets, ensuring that reasoning performance is preserved without allowing memory to grow indefinitely [55]. Complementing these forgetting mechanisms, biologically inspired architectures like FadeMem implement differential decay rates across dual-layer hierarchies, using adaptive exponential decay functions modulated by semantic relevance to allow irrelevant details to fade while consolidating related information [291]. Similarly, the MARS framework leverages the Ebbinghaus forgetting curve to optimize information storage and transmission, enabling agents to adaptively adjust strategies and reduce cognitive load during multi-tasking scenarios [252].

The stabilization of learning signals for these adaptive decisions often relies on sophisticated reinforcement learning techniques. Step-wise Group Relative Policy Optimization (GRPO) has emerged as a critical tool for training agents to master the timing of compression and retrieval decisions. For instance, the InfoMem framework utilizes answer-conditioned information gain within a GRPO setting to evaluate final-memory utility, stabilizing optimization by applying rewards only to successful trajectories [72]. This approach addresses the sparsity of rewards in long-horizon workflows, a challenge also tackled by ZipRL, which combines multi-granularity compression with Hindsight Response Replay to densify training signals and balance information retention with token efficiency [268]. Furthermore, the Agentic Memory (AgeMem) framework employs step-wise GRPO to unify long-term and short-term memory management directly into the agent's policy, enabling autonomous decisions on when to store, retrieve, or discard information [10].

Compression-adaptive training and dynamic context management are also pivotal in ensuring that salient information remains accessible. StreamingDialogue introduces the concept of "conversational attention sinks" to compress long dialogue history into End-of-Utterance tokens, reducing computational complexity quadratically while employing short-memory reconstruction and long-memory reactivation strategies to minimize information loss [279]. In parallel, the ProStream framework utilizes Adaptive Spatiotemporal Optimization to dynamically optimize retention based on expected utility, enabling bounded knowledge states that support ad-hoc memory recall in infinite-horizon streaming dialogues [153]. These methods highlight a shift from passive context extension to active, learned management of memory states.

Discrepancies exist regarding the optimal granularity of memory operations. While some systems advocate for lightweight construction paired with sophisticated utilization to reduce overhead [136], others propose dual-process systems that bridge lightweight vector retrieval with exhaustive deliberation modules via multi-dimensional quality gating [293]. The MemMachine system further refines this by demonstrating that retrieval-stage optimizations, such as depth tuning and query bias correction, often yield greater accuracy improvements than ingestion-stage changes, suggesting that adaptive retrieval strategies are more critical than raw storage fidelity in certain contexts [147]. Additionally, hierarchical graph-based approaches like GAM and HiMem decouple memory encoding from consolidation to resolve conflicts between rapid context perception and stable knowledge retention, employing conflict-aware reconsolidation to revise stored knowledge based on retrieval feedback [41] [33].

The integration of these mechanisms ensures that agents can operate effectively within bounded context budgets while improving overall retention rates. Techniques such as the Multi-dimensional Quality Gating policy in D-Mem have been shown to recover nearly all the performance of full deliberation modules at a fraction of the computational cost, validating the efficacy of dynamic gating [293]. Moreover, the adoption of retention regularization in multi-layer frameworks has proven effective in reducing false memory rates to as low as 5.1% while maintaining high multi-hop reasoning capabilities [5]. Collectively, these advancements illustrate a consensus that adaptive gating, coupled with rigorous retention regularization and importance-weighted retrieval, is essential for mitigating semantic drift and enabling robust, long-term agent performance.

Indexed and Externalized Experience Memory

A primary strategy for overcoming the finite context windows of Large Language Model (LLM) agents involves the externalization of full-fidelity interaction histories into indexed databases while maintaining a compact, indexed summary within the working context. This architectural pattern effectively decouples the size of the experience archive from the constraints of the prompt, treating memory as an external storage tier where detailed artifacts are stored under stable indices and retrieved on demand. The foundational work in this domain, [7], introduces a mechanism where the agent compresses context without discarding evidence by replacing long tool-use trajectories with concise structured summaries and stable indices. In this framework, the full underlying interactions are archived in an external key-value store, allowing the agent to explicitly dereference an index to recover exact past evidence when relevant to a current subgoal. This approach contrasts sharply with lossy summarization or truncation methods, offering a theoretically bounded in-context computation that preserves decision quality even as history grows. The efficacy of this indexed approach is further supported by [63], which proposes a semantic lossless compression pipeline that distills unstructured interactions into compact, multi-view indexed memory units, achieving significant reductions in token consumption while improving retrieval efficiency.

The distinction between maintaining a full context window versus utilizing an externalized memory system is not merely architectural but also economic and performance-based. Comparative analyses reveal a nuanced trade-off: while long-context models may offer higher factual recall on specific benchmarks, external memory systems often provide a superior cost-performance profile for extended interactions. [2] demonstrates that while long-context inference incurs per-turn charges that grow with context length, memory systems maintain a roughly fixed read cost after an initial write phase, becoming cheaper after approximately ten interaction turns at high token counts. However, this cost efficiency does not always guarantee superior accuracy across all tasks; the same study notes that long-context models can outperform memory systems on factual recall benchmarks like LongMemEval, suggesting that the choice between architectures depends heavily on the specific requirements for persona consistency versus raw fact retrieval. Furthermore, [204] highlights that no single memory form works consistently across all settings, noting that retrieval-based methods remain strong for knowledge-intensive tasks, whereas procedural and long-term memory methods are more effective for execution-oriented tasks when the stored experience matches the task structure.

To optimize the utility of these externalized memories, recent research has focused on learning when and how to manage the index and retrieval process actively. [7] employs a reinforcement learning framework, MemexRL, to optimize both write and read behaviors, teaching the agent what to summarize, how to index, and when to retrieve based on reward shaping tailored to indexed memory usage. Similarly, [158] formulates context management as a sequential decision problem with a budget constraint, using reinforcement learning to determine when and how much history to compress. This active management is crucial because passive accumulation or static retrieval often leads to "context rot" or performance degradation. [70] argues that context must be treated as a dynamic internal reasoning state, proposing a reflection-driven process that actively reorganizes working context when misalignment is detected, outperforming passive compression methods. The concept of active management extends to the structure of the memory itself; [18] organizes interaction traces into a graph of reasoning steps, selecting context based on logical dependencies rather than simple temporal proximity, thereby preserving the causal structure necessary for multi-step reasoning.

Despite the advantages of indexed external memory, challenges remain regarding the fidelity of retrieval and the handling of dynamic information updates. [43] introduces a benchmark to evaluate whether memory systems can help agents acquire environment-specific experience, revealing that while coding-agent-based methods achieve high accuracy, they suffer from high latency costs. Moreover, the static nature of many retrieval systems struggles with evolving information. [40] proposes a causal memory-selection technique that estimates how candidate memories affect the model's answer, aiming to suppress unstable or misleading memories that semantic similarity might incorrectly retrieve. This is particularly important in environments where facts change over time, a gap highlighted by [44], which shows that current systems struggle to recall and reason over earlier facts that are later revised or interfered with by subsequent context. To address the fragmentation of information in external stores, [160] proposes a three-tiered directory driven by user-centric relevance, promoting records mapped to long-term patterns toward active tiers while relegating less relevant information, thus balancing lean active search spaces with comprehensive archives.

The evolution of these systems suggests a shift from simple retrieval-augmented generation toward more sophisticated, state-adaptive, and hierarchical architectures. [20] consolidates ongoing interactions into compact memory cues while preserving raw trajectory pages for intent-driven recall, treating cues as lightweight handles to reconstruct temporally distant information. Similarly, [36] organizes conversations through a Temporal Memory Tree, enabling systematic consolidation from raw observations to abstracted persona representations. These hierarchical and state-adaptive approaches aim to resolve the tension between the need for compact working contexts and the necessity of preserving fine-grained evidence, a tension that [58] addresses through "folding" operations that manage historical trajectories at multiple scales, performing granular condensations for vital details and deep consolidations for entire sub-tasks. Ultimately, the literature indicates that the most robust agents will likely employ hybrid strategies that combine the cost-efficiency of externalized indexed memory with the high-fidelity recall capabilities of learned, active management policies, ensuring that the decoupling of archive size from context window does not come at the expense of reasoning coherence or temporal awareness.

Indexed Summaries and Explicit Dereferencing

The core challenge in scaling Large Language Model (LLM) agents to infinite-horizon tasks lies in the fundamental tension between the finite capacity of context windows and the unbounded growth of interaction trajectories. Traditional approaches that rely on truncation or lossy summarization often discard critical evidence, leading to degraded reasoning quality as tasks extend over time. To address this, a paradigm shift has emerged toward indexed memory systems that decouple the working context from the archival store. The seminal work [7] introduces a mechanism where agents generate concise, structured summaries that act as pointers to detailed external records. This approach allows the agent to reason over a small, bounded context while retaining the ability to reconstruct full historical trajectories through explicit dereferencing when specific evidence is required. By treating the file system or external index as the primary source of truth, [7] demonstrates that it is possible to achieve bounded in-context computation without sacrificing decision quality, effectively solving the context overflow problem while preserving exact evidence recovery.

This concept of explicit dereferencing finds strong theoretical and practical support across various contemporary frameworks that treat memory not merely as storage but as an executable state management layer. For instance, [114] parallels this philosophy by externalizing persistent state into a file-centric abstraction, allowing the agent to reconstruct its context from a workspace snapshot and a fixed window of recent actions. Similarly, [31] argues that semantic similarity retrieval is insufficient for execution-state dependencies, proposing instead a hierarchical state tree where agents explicitly compress completed subgoals and revise boundaries to isolate flawed segments. These systems collectively suggest that the fidelity of long-horizon reasoning depends on the agent's ability to distinguish between high-level progress indicators (summaries) and low-level evidentiary data (archives), retrieving the latter only when necessary for specific subgoals.

The mechanism of indexed dereferencing also addresses the "fidelity dilemma" identified in ultra-long context processing, where aggressive compression risks erasing subtle bridging facts required for multi-hop reasoning. [8] highlights that passive memory updates often fail to preserve low-salience evidence, necessitating active control protocols that monitor evidence sufficiency and perform targeted retrieval. This aligns with the findings in [75], which warns that continuous consolidation of trajectories into abstracted text can introduce errors and overwrite ground-truth evidence. Consequently, systems like [7] and [63] advocate for retaining raw episodes as first-class evidence, using indexed summaries only as navigational aids rather than replacements for the original data. [63] further refines this by employing semantic lossless compression to distill interactions into indexed units, ensuring that the retrieval process remains efficient without incurring the information loss associated with generative summarization.

Furthermore, the efficacy of indexed memory systems is enhanced when the dereferencing process is learned rather than heuristic. [7] utilizes a reinforcement learning framework, MemexRL, to optimize both the writing of indexed summaries and the timing of dereferencing actions, treating memory operations as first-class decisions within the agent's policy space. This learning-based approach contrasts with static retrieval methods and resonates with [13], which formulates context management as learnable in-place editing operations. By training agents to recognize when a summary is insufficient and an index must be dereferenced, these systems mitigate the risk of "context rot" described in [70], where passive accumulation leads to incoherent internal states. The integration of such learned policies ensures that the agent maintains a compact working state while possessing the procedural knowledge to expand its context precisely when the task demands high-fidelity evidence.

The architectural implications of indexed dereferencing extend to the organization of memory itself, moving away from flat vector stores toward hierarchical and graph-based structures that facilitate precise navigation. [284] replaces flat global summaries with time-ordered trees, enabling localized updates and efficient retrieval paths that mirror the indexed summary concept. Similarly, [217] employs dynamic indexing and linking inspired by the Zettelkasten method to create interconnected knowledge networks, allowing agents to traverse from high-level concepts to specific historical notes. These structural innovations support the theoretical claim that bounded dereferencing can preserve decision quality even as history grows indefinitely, a proposition central to the design of [7]. By combining structured indexing with learned retrieval policies, modern agent architectures are increasingly capable of sustaining coherent, evidence-based reasoning over horizons that would otherwise exceed the limits of standard context windows.

Dependency-Structured Memory Graphs

The evolution of agent memory from passive storage to active cognitive scaffolding marks a critical shift in addressing the limitations of flat, similarity-based retrieval. Traditional Retrieval-Augmented Generation (RAG) systems often treat memory as a stateless lookup table, retrieving isolated facts based on semantic proximity while neglecting the causal and logical dependencies required for multi-step reasoning. This structural mismatch frequently leads to context fragmentation, where agents retrieve relevant keywords but fail to reconstruct the execution state or reasoning trajectory necessary to solve complex problems. To overcome these deficiencies, recent research has pivoted toward dependency-structured memory graphs that explicitly model the relationships between reasoning steps, ensuring that retrieved context preserves the integrity of causal chains.

A primary advancement in this domain is the transition from "Memory-as-Tool" to "Memory-as-Cognition," where memory access becomes an integral part of the reasoning process rather than a peripheral query. [193] argues that standard one-shot retrieval suffers from reasoning-retrieval decoupling, proposing instead a navigable memory store with associative link graphs that support multi-step traversal. Similarly, [18] introduces a framework specifically designed to organize interaction traces into graphs of reasoning steps. Unlike sliding window approaches that discard early structured information, ContextWeaver utilizes dependency-based construction to link each step to its predecessors, allowing agents to select context based on logical reliance rather than mere semantic similarity. This approach has proven particularly effective in software engineering tasks, where complex logical traversal is essential, significantly reducing reasoning steps and token usage while improving pass rates on benchmarks like SWE-Bench Verified.

The necessity of modeling logical dependencies is further reinforced by architectures that decouple memory representation from retrieval logic to enhance interpretability. [185] addresses the entanglement of temporal, causal, and entity information in monolithic stores by representing memory items across orthogonal semantic, temporal, causal, and entity graphs. By formulating retrieval as a policy-guided traversal over these relational views, MAGMA enables query-adaptive selection that constructs structured contexts aligned with the agent's specific navigational needs. This contrasts with systems that rely solely on dense vector embeddings, which often obscure the fine-grained details required for effective reasoning. Complementing this, [198] transforms unstructured dialogue history into a structured causal and semantic graph, leveraging counterfactual reasoning to deduce implicit constraints and resolve conflicts between past states and current intentions.

Furthermore, the concept of active reconstruction suggests that memory should not be statically retrieved but dynamically assembled based on intermediate evidence. [294] proposes MRAgent, which combines an associative memory graph with an active reconstruction mechanism. This framework allows agents to iteratively explore and prune retrieval paths based on accumulated evidence, preventing the combinatorial explosion of unconstrained expansion while ensuring memory access adapts to the evolving reasoning context. This dynamic adaptation is crucial for long-horizon tasks where the relevance of specific memory fragments changes as the agent progresses through a workflow. [34] extends this by constructing a dependency-aware memory that captures salient intermediate states and their logical relations, actively pruning invalid steps and folding completed sub-trajectories to maintain a compact, high-salience reasoning backbone.

Despite the consensus on the value of structured dependencies, various implementations offer distinct mechanisms for organizing these graphs. [217] employs principles from the Zettelkasten method to create interconnected knowledge networks through dynamic indexing, allowing new memories to trigger updates in historical contextual representations. In contrast, [31] argues that semantic similarity mismatches execution-state dependencies, proposing MAGE, which stores interactions in a hierarchical state tree to isolate flawed segments and preserve state integrity. Meanwhile, [67] models memory as a dynamic graph where relevance emerges from spreading activation rather than pre-computed links, integrating lateral inhibition to filter interference. These diverse approaches highlight a growing recognition that effective long-horizon reasoning requires memory systems that do not just store information, but structurally encode the dependencies that give that information meaning. By shifting focus from isolated fact retrieval to the reconstruction of dependency graphs, these frameworks ensure that agents can navigate complex problem spaces with the coherence and logical continuity previously limited to human cognition.

Causality and Tool-Augmented Retrieval

Traditional retrieval-augmented generation (RAG) systems often rely on semantic similarity to retrieve memories, an approach that frequently fails to capture the objective causality required for complex agent-environment interactions. This limitation is particularly acute in long-horizon tasks where the temporal and logical dependencies between actions determine the validity of future states. To address the fragility of similarity-based retrieval, which can retrieve topically related but causally irrelevant or even misleading information, recent research has pivoted toward integrating causality graphs and tool-augmented retrieval mechanisms. These advanced architectures ensure that retrieved memories are not merely semantically adjacent but are structurally linked to the current agent state, thereby improving accuracy in benchmarks testing for objective information retention and logical consistency [40]. By deriving state from active paths and isolating flawed segments, these systems preserve state integrity and prevent the cascading errors that plague monolithic memory stores [31].

A primary strategy in this domain involves the decoupling of memory representation from retrieval logic through multi-graph architectures. Systems like MAGMA represent memory items across orthogonal semantic, temporal, causal, and entity graphs, formulating retrieval as a policy-guided traversal over these relational views. This allows for query-adaptive selection and the construction of structured contexts that align with the specific intent of the agent, rather than relying on dense vector embeddings that entangle distinct types of information [185]. Similarly, ActMem transforms unstructured dialogue history into a structured causal and semantic graph, leveraging counterfactual reasoning and commonsense completion to deduce implicit constraints. This approach enables agents to resolve conflicts between past states and current intentions, moving beyond passive recording to active causal reasoning [198]. The integration of such causal structures is critical because, as noted in evaluations of long-horizon agents, existing systems often underperform due to a lack of causality and an over-reliance on lossy similarity metrics [17].

Tool-augmented retrieval further enhances these capabilities by treating memory access as an executable action rather than a passive lookup. In frameworks like MRAgent, memory is represented as a Cue-Tag-Content graph where an active reconstruction mechanism integrates LLM reasoning directly into the access process. This allows the agent to iteratively explore and prune retrieval paths based on accumulated evidence, dynamically adapting to the reasoning context while avoiding combinatorial explosion [294]. This active engagement with memory is echoed in MemCog, which shifts from a "Memory-as-Tool" paradigm to "Memory-as-Cognition," exposing navigable interfaces that allow agents to perform multi-step reasoning-driven traversal across associative link graphs [193]. By enabling agents to actively reconstruct context rather than simply retrieving static fragments, these systems mitigate the risk of retrieving stale or contradictory information that can derail long-term planning.

The necessity of causal grounding is further highlighted by the failure modes observed in systems that continuously update consolidated memories without rigorous validation. Research indicates that useful memories can become faulty when continuously updated by LLMs, as abstraction steps may introduce spurious rules or drop critical details, leading to performance degradation below no-memory baselines [75]. To counteract this, approaches like Causal Memory Intervention (CMI) estimate how candidate memories affect the model's answer under controlled interventions, selecting only those that improve task performance while suppressing unstable or harmful entries. This method demonstrates that reliable long-term memory requires selecting context based on causal usefulness rather than relevance alone, a finding supported by the introduction of causally annotated benchmarks like CAUSAL-LOCOMO [40].

Furthermore, the management of execution state is paramount for preventing error cascades in interdependent decision trajectories. MAGE (Memory as Agent-Guided Exploration) addresses this by storing interactions in a hierarchical state tree, where the agent derives its state strictly from the active root-to-current path. This design isolates flawed segments from the active path, allowing the agent to revise boundaries and resume on new branches without contaminating the current context with erroneous traces [31]. Complementing this, ContextWeaver organizes interaction traces into a graph of reasoning steps with explicit dependency links, ensuring that each step is connected to the earlier steps it relies upon. This dependency-based construction supports compact summarization and validation, providing a stable mechanism for agents that utilize tools in complex, multi-step environments [18].

The shift toward causality and tool augmentation also addresses the "governance gap" in autonomous agents, where standard RAG lacks the controls to manage contradictions or prevent outdated facts from polluting the context. By

enforcing explicit, rule-based policies and utilizing graph structures to track state evolution, systems like MemArchitect and those employing Governed Evolving Memory (GEM) ensure that correctness is a property of the state trajectory rather than individual records [69; 168]. These advancements collectively suggest that the future of agentic memory lies not in expanding context windows or refining embedding spaces, but in constructing explicit, traversable causal models that allow agents to reason about the consequences of their past actions and the validity of their retrieved knowledge.

Cost-Performance Trade-offs in External Memory

The deployment of persistent conversational agents necessitates a rigorous evaluation of the economic and performance implications associated with context management strategies. A fundamental tension exists between leveraging expanding native context windows in large language models (LLMs) and implementing externalized memory systems that offload historical data. Comparative analyses indicate that while long-context models demonstrate superior factual recall on specific benchmarks, external memory architectures offer a distinctly advantageous cost-performance profile for extended interaction horizons. This economic divergence arises from the structural differences in how inference costs accumulate: long-context inference incurs a per-turn charge that scales linearly or quadratically with the growing token count, whereas external memory systems typically involve a fixed per-turn read cost following an initial write phase [2]. Consequently, for contexts exceeding 100,000 tokens, indexed and fact-based memory systems become significantly more cost-effective after a modest number of interaction turns, often reaching a break-even point within ten turns as the context length increases.

The limitations of relying solely on native context extension are multifaceted, extending beyond mere financial cost to include cognitive degradation and computational inefficiency. As interaction histories grow, models frequently suffer from the "lost-in-the-middle" phenomenon, where attention dilution causes performance to degrade despite the presence of relevant information within the window [32]. Furthermore, the quadratic complexity of self-attention mechanisms in standard transformer architectures means that processing ever-lengthening histories incurs prohibitive latency and memory overhead, rendering full-context approaches unsustainable for long-horizon tasks [290]. Even when models possess million-token context windows, empirical evidence suggests that their effective utilization of such vast contexts remains suboptimal, with performance dropping dramatically as context length increases due to noise accumulation and attention saturation [37]. In contrast, external memory systems decouple the storage of evidence from the working context, allowing agents to maintain a compact, high-density state while preserving full-fidelity historical data in external databases [7].

Architectural innovations in external memory have evolved to address the specific trade-offs between retrieval accuracy, token efficiency, and computational cost. Systems like MemGPT and MemOS draw inspiration from operating system hierarchies, implementing virtual context management that pages data between fast working memory and slower archival storage, thereby creating the illusion of infinite context without the associated inference costs [290] [109]. More advanced frameworks, such as Memex, introduce indexed experience memory where agents learn via reinforcement learning to compress trajectories into structured summaries while archiving raw evidence under stable indices, enabling precise dereferencing only when necessary [7]. This approach mitigates the lossiness inherent in simple summarization techniques, which often discard critical details required for complex reasoning [30]. Similarly, methods like SimpleMem and CoM (Chain-of-Memory) focus on semantic compression and dynamic evolution to maximize information density, achieving substantial reductions in token consumption-up to 30 times less than baseline approaches-while maintaining or improving accuracy on long-term memory benchmarks [63] [136].

However, the transition to external memory is not without its challenges, particularly regarding the reliability of memory consolidation and the complexity of retrieval mechanisms. Research indicates that continuous updates to consolidated memory by LLMs can introduce faults, where useful abstractions degrade over time, sometimes performing worse than retaining raw episodic traces [75]. This suggests that while external memory offers cost advantages, the strategy for writing and updating memory must be carefully gated to prevent the accumulation of erroneous abstractions. Furthermore, retrieval-based systems face the risk of retrieving irrelevant or misleading information if not grounded in causal relationships rather than mere semantic similarity [40]. To address these issues, recent works propose dual-process architectures that combine lightweight vector retrieval for routine queries with high-fidelity deliberation modules for complex tasks, optimizing the balance between speed and accuracy [293]. Other approaches, such as PRISM and MemForest, utilize graph-structured memory and hierarchical temporal indexing to reduce maintenance overhead and improve retrieval precision under strict context budgets [280] [284].

The viability of external memory systems for production deployments is further reinforced by their ability to handle domain-specific requirements that native context windows struggle to meet. In scientific workflows, for instance, where parameter evolution and contradictory updates are common, dual-process architectures separating episodic buffers from semantic consolidation have demonstrated the ability to maintain accuracy where full-context models fail due to

overflow [11]. Additionally, the economic argument is strengthened by findings that optimized retrieval strategies can yield higher accuracy than larger, more expensive models when paired with efficient prompt designs, as seen in systems like MemMachine which achieve superior results with significantly fewer input tokens [147]. Ultimately, the choice between long-context inference and external memory is not binary but depends on the interaction horizon and the specific cost constraints of the deployment. For applications requiring extensive interaction histories exceeding 100k tokens, the fixed-cost structure of external memory systems provides a scalable and economically sustainable pathway, provided that the mechanisms for memory writing, consolidation, and retrieval are robust enough to preserve the integrity of the agent's evolving knowledge base [2] [239].

Conclusion

The literature reveals a definitive paradigm shift in the architecture of Large Language Model agents, moving away from static, passive retrieval mechanisms toward dynamic, active, and learnable memory management systems. The fundamental limitation of relying solely on expanding context windows or standard Retrieval-Augmented Generation has become apparent: these approaches suffer from attention dilution, the "lost in the middle" phenomenon, and an inability to resolve causal contradictions or manage state evolution over long horizons. Consequently, the field is converging on continuum memory architectures that treat memory not as an external database but as an integral, executable component of the agent's policy. This transition enables agents to actively curate their cognitive state through learned operations such as compression, consolidation, and selective forgetting, thereby maintaining high information density and reasoning coherence despite finite context constraints.

A central theme across diverse research efforts is the necessity of structural sophistication to preserve logical integrity. Monolithic storage is increasingly being replaced by hierarchical, multi-layered, and graph-based architectures that decouple transient episodic traces from stable semantic knowledge. By organizing memory into distinct layers-ranging from working memory for immediate subgoals to indexed archives for full-fidelity evidence-these systems effectively balance the competing demands of efficiency and fidelity. The emergence of indexed experience memory and explicit dereferencing mechanisms allows agents to operate within bounded context windows while retaining the ability to reconstruct precise historical trajectories on demand. Furthermore, the integration of dependency-structured graphs and causal reasoning modules addresses the fragmentation inherent in flat retrieval, ensuring that agents can navigate complex, multi-hop reasoning paths without losing the causal chains that link past actions to current states.

Despite these architectural advances, significant challenges remain regarding the reliability of memory consolidation and the optimization of learning signals. A critical tension exists between the need for abstraction to scale and the risk of introducing faults through continuous updates; research indicates that aggressive summarization can degrade performance below baseline levels if not carefully gated. This underscores the importance of treating raw episodic evidence as first-class data and employing biologically inspired forgetting mechanisms to prevent interference. Moreover, the development of robust reinforcement learning strategies is essential to solve the sparse reward and credit assignment problems inherent in long-horizon tasks. Future progress depends on the synthesis of these elements into unified frameworks that can dynamically adapt memory strategies to specific domain constraints, supported by rigorous evaluation benchmarks that assess causality, logical consistency, and interdependent task performance rather than static recall alone. Ultimately, the path toward generalizable agentic memory lies in creating self-evolving systems that govern their own lifecycle, balancing retention, forgetting, and adaptation to achieve true lifelong learning.